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A Novel lightweight Stream Cipher based on NLFSRs and Chaos function

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Abstract— **The Internet of Things (IoT) is already an established phenomenon where numerous devices are interconnected and used daily. Such devices collect and send out enormous amounts of information, and experts believe people will depend more on IoT over the years. As the volume of shared information continues to grow, there is an increasing need to strengthen security measures. The wide range of data IoT must protect is difficult due to IoT's lack of capacity regarding hardware, power, and storage. Conventional encryption techniques cannot be utilized effectively in these types of surroundings. To solve these obstacles, we suggest a lightweight cipher capable of strong encryption without abundant resources. In this study, Two Non-Linear Feedback Shift Registers (NLFSRs) are utilized in conjunction with a logistic chaotic function where the NLFSRs are initialized using 40-bit seed keys. Our method, in comparison to the best algorithms such as Trivium, Grain, and Grain128 performs the best when it comes to the speed and number of encryption and decryption cycles that can be completed within a certain duration. In addition to proving compliance with NIST, the algorithm achieves an impressive strong avalanche effect where even minimal alterations to the input result in substantial and unpredictable results. This feature, together with the high level of increase in the keystream and the low level of security the keystream offers, increases the algorithm's resistance to cryptographic attacks.**

Keywords— Keystream, Logistic chaos function, NLFSR, NIST Stream Cipher.

I. INTRODUCTION

The Internet of Things (IoT) describes a network in which various devices and objects are connected through the Internet, allowing them to communicate using either wireless or wired connections. The Internet of Things has seen a sharp increase in popularity due to its many uses in corporate development, education, transportation, and communication[1]. The ease of connecting devices to a network brings in many innovative ideas and possibilities across multiple industries. The rapid diffusion of the IoT phenomenon has greatly aided organizations in improving market research and corporate strategy formulation. Automated services have changed people's lifestyles for the better. But along with uncontrolled IoT expansion comes heightened security and privacy concerns. The deployment of IoT technologies needs to be approached thoughtfully and comprehensively to eliminate or reduce the threats brought about by interconnectivity. With interconnected devices, the amount of information exchanged is staggering, and therefore, one can never be too sure if a breach is truly harmless. IoT describes a network in which various devices and objects are connected through the Internet, allowing them to communicate using either wireless or wired connections. In the case of the IoT, novel encryption approaches must be energy efficient and low in memory usage. Moreover, incorporating such approaches into hardware should be straightforward. Out of all the many security factors, the most attractive one that still captures the interest of numerous specialists is how to achieve the optimum level of computational cost against the level of protection in the context of constrained resources.

Asymmetric cryptographic algorithms are generally not recommended for resource-constrained environments because they require significant computational power, memory, and energy consumption. Thus, there has been a great deal of interest in lightweight symmetric cryptography [24]. The main reason for this emphasis is its capacity to deliver sufficient security in IoT devices. Such smart and efficient security measures and policies enable protection for IoT networks and devices, complementing the domain of symmetric lightweight cryptography. Symmetric ciphers are generally classified into two main categories: stream ciphers and block ciphers. Numerous instances of both block and stream cipher methods have been shown such as AES, DES, Grain, Grain 128, and Trivium [25]. The internal state sizes of Grain, Grain-128, and Trivium are 160 bits, 256 bits, and 288 bits, respectively. Grain and Grain-128 are constructed using a combination of LFSR and NLFSR. While Trivium uses three Non-linear shift registers. Because of their linearity, the Grain and Grain-128 ciphers are very attack-prone. It is easy for an adversary to reverse the keystream generation scheme to obtain the key. This paper uses a hub with a combined logistic chaos function and a nonlinear feedback shift register to create secure, lightweight ciphers [26-28].

In this research, two NLFSRs with 40-bit keys and a logistic function were used. The logistic function serves as a seed for the 40-bit keys in the NLFSRs. The mixture of the non-linear

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logistic chaos function and non-linear FSR produces an extremely unpredictable keystream, which makes the cipher exceedingly difficult to break [29].

Our cipher consists of 80-bit keys, which use very little memory compared to state-of-the-art algorithms.

To better illustrate this work's significance, we have enumerated the benefits of the proposed solution below.

The logistic chaos function is used to initialize the 80-bit NLFSRs key, and it is also nonlinear.

- The two 40-bit NLFSR's produce a keystream that has a large degree of randomness which greatly increases the complexity of attempting to compromise the secret keys or to successfully recover them in the event of an attack.
- This method is highly secure; it requires minimal memory and produces high-throughput rates.

Other parts of this literature are presented in the following way: Section II explores related work. Section III talks about design specifications. Sections V and VI expound on the proposed work and report results and discussions. Section VI presents the observations, while the conclusions and future directions are addressed in Section VII.

II. Related work

Classic Advanced Encryption Standard (AES) block encryption remains notoriously cumbersome due to multiple iterations over a round function, resulting in a higher consumption of both time and resources. There is still work to do in alleviating the shortcomings of cipher: for example, Ho's work with Hummingbird2 suggests that a minimum of 4 rounds is needed with most classic block cyphers. Although stream cyphers have a greater speed, operating memory, and processing power when compared to block cyphers, their efficiency is still limited. There are many proposals for lightweight cyphers such as Grain and Grain-128; however, these constructions remain problematic. These systems are often highly susceptible to analytical attacks because a round function with fixed cryptographic primitives and a variable number of rounds r is employed.

Chaotic cryptography is the foundation of another similar strategy. Nevertheless, there are various performance and security issues with the suggested techniques. Because they need conversion operations and floating-point calculations, they are susceptible to some attacks and have performance issues. Additionally, as seen in [8]These cyphers use a multi-round structure. The IoT appliances operate in a multitasking mode, meaning they serve as sensors and transmitters. Furthermore, fast throughput and lightweight methods should be employed to provide security. For instance, an Internet of Things application in a distant forest or on a farm could not have access to a charging station. In [9] et al. the author proposes three key tactics to address the problems above. The other area of study is the creation of lightweight stream ciphers appropriate for environments with constraints. In [10], The limitations of block ciphers were addressed by evaluating the effectiveness of stream ciphers in wireless network environments. As well as in [11], the author performed a security and performance scrutiny of stream ciphers for low-end embedded devices. In the design of certain stream ciphers, the internal state size must reach or exceed a specified threshold to ensure adequate security. More recent work has been done to adapt an LFSR-based stream cipher by putting a cap on the internal state size while also working with a low-cost resource constrained setting [12][13][14]. Sprout [13], Fruit [15], Lizard [16], and Plantlet [17] are among many other well-known stream ciphers that were results of these efforts. Several studies [18][19] have reported that these ciphers are highly susceptible to a wide range of advanced attacks, such as time-memory, fault, distinguishing and other attacks. Such attacks are termed advanced because they are relatively more intensive in terms of resources, processing power, and memory demand. Considering these parameters frequently gets ignored when securing networks of low-cost devices. Rather more reserved measures like statistical tests, which fall short of providing reliable assessments, must be used when evaluating the security of lightweight cryptography [20][21].

III. Design specification

A lightweight stream cypher implementation that makes use of the logistic chaos function and NLFSR has already been presented in this research. All-purpose hardware-oriented cypher works on devices that are memory and power-constrained, owing to its lightweight resource-oriented design. Our proposed work's structure is like the grain cypher, which is depicted in Figure 1. In contrast, the proposed method uses two 40-bit NLFSRs, employs a chaotic function and uses a filter function that has five variables of algebraic degree 4 to improve nonlinearity in the cipher. Figure 1 shows the complete architecture of the proposed system.

The update operation for NLFSR-1 is defined as follows.

$$\delta_{i+40} = \omega_i + \delta_i + \delta_{i+12} + \delta_{i+17} + \delta_{i+21} + \delta_{i+25} + \delta_{i+6}\,\delta_{i+29} + \delta_{i+20}\,\delta_{i+11} + \delta_{i+11}\,\delta_{i+3} + \delta_{i+23}\,\delta_{i+22}\,\delta_{i+14} + \delta_{i+18}\,\delta_{i+12}\,\delta_{i+5} + \delta_{i+10}\,\delta_{i+16}\,\delta_{i+3} + \delta_{i+22}\,\delta_{i+13}\,\delta_{i+9}\,\delta_{i+4} \quad (1)$$

And the update operation for NLFSR-2 is defined as follows.

$$\omega_{i+40} = \omega_i + \omega_{i+22} + \omega_{i+24} + + \omega_{i+19} + \omega_{i+5} + \omega_{i+18}\,\omega_{i+21} + \omega_{i+25}\,\omega_{i+17} + \omega_{i+19}\,\omega_{i+3} + \omega_{i+29}\,\omega_{i+17}\,\omega_{i+5} + \omega_{i+21}\,\omega_{i+14}\,\omega_{i+7} + \omega_{i+18}\,\omega_{i+17}\,\omega_{i+8}\,\omega_{i+3} + \omega_{i+25}\,\omega_{i+14}\,\omega_{i+11}\,\omega_{i+5} \quad (2)$$

As described below, the function attains a maximum nonlinearity value of 12. The filter function employed in the proposed scheme is a balanced Boolean function with a nonlinearity of 12, defined over five variables and having an algebraic degree of four.

$$g(z) = z_1 \oplus z_3 \oplus z_1z_4 \oplus z_1z_5 \oplus z_2z_3 \oplus z_2z_5 \oplus z_3z_4 \oplus z_1z_2z_3 \oplus z_1z_2z_3z_4 \oplus z_1z_2z_3z_5 \oplus z_1z_2z_4z_5 \oplus z_1z_3z_4z_5 \oplus z_2z_3z_4z_5 \quad (3)$$

The values of the variables z_1, z_2, z_3, z_4 and z_5 correspond to state bits. ω_{i+2} , ω_{i+17} , ω_{i+29} , δ_{i+32} and δ_{i+39} The definition of the output function is

$$= \sum_{k\in C} \omega_{i+k} \oplus g(z) \quad (4)$$

where $C \in [3,4,7,15,23,28,34,37]$

A. Components

The two components that we wish to focus on are the Logistic chaos function and the two NLFSRs.

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A.1 Logistic Chaos Function

Numerous investigations have been conducted regarding chaos theory in relation to its application within nonlinear science and cryptography. These traits are very much akin to the current concepts of cryptographic confusion and diffusion at [22]. Chaotic systems exhibit several properties that are advantageous for cryptographic applications, including high sensitivity to initial conditions, pseudo-random behavior, and long-period oscillations. The effect of finite precision is mitigated by introducing a method of chaotic digitization. The logistic chaotic map expression is displayed as

$$y(t+1) = \gamma y(t)\{1 - y(t)\} \quad (5)$$

where $y_t$represents the state variable at iteration t, $\gamma$is the control parameter satisfying $3.6 < \gamma < 4$, and $y_t \in (0,1]$.

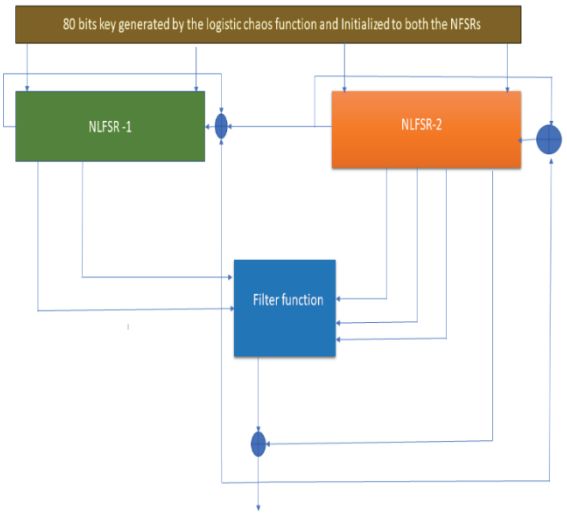


Fig. 1 Architecture of Proposed Work

A.2 Non-Linear Feedback Shift Register (NLFSR)

This subsection employs two NLFSRs, where NLFSR-2 provides feedback to the LSB of NLFSR-1. Both registers operate in parallel under the same clock signal. The first value of the chaotic sequence is initialized to prevent a zero state. Once initialization is completed, the clock advances, updating the states of NLFSR-1 and NLFSR-2 [23].

IV. Proposed Work

Our suggested approach uses an 80-bit key initially set to two 40-bit NFSRs and generated randomly using the logistic chaos function. In our study, we use the state updating function for each NFSR given in equations 2 and 3 to generate the maximum random key. We also utilize a clock function, clocking 80 times to intermix the bits of two The NLFSRs are clocked for 80 cycles without generating any output. After these 80 clocking cycles, the 81st key is utilized for the encryption and decryption processes. The encryption and decryption algorithms of the proposed scheme are described in this section.

A. Encryption

Input: Plain text P
Output: Cipher

1. y_0 = random (0,1)
2. γ = random(3.6, 4)
3. for i = 0 to 80
   $g[]$= logistic_function(y_0, γ)
   # initialize the first 40 bits in nfsr-1
4. $NFSR_1[]$= $g[:40]$
   # initialize the remaining 40 bits in nfsr-2
5. $NFSR_2[]$= $g[40:]$
6. clock ()
7. stream = stream_generation()
8. cipher = $P \oplus$ **Stream**
9. Return cipher

B. Decryption

Input: Cipher , γ, y_0
Output: Plain text

1. for i = 0 to 80
   $g[]$= logistic_function(y_0, γ)
2. $NFSR_1[]$= $g[:40]$
3. $NFSR_2[]$= $g[40:]$
4. clock()
5. stream = stream_generation()
6. Plain_text = *Cipher* $\oplus$ **Stream**
7. return Plain_text

V. Result and Discussion

Experiments were performed to evaluate data from the very well-defined 20 Newsgroups data set, which has been extensively used in many previous studies. The experiment was conducted on a personal computer that had an Intel Core i7 as well as 16 GB of RAM, a 256 GB SSD, and Windows 11 installed. The proposed project has been compared with several state-of-the-art algorithms such Grain improved 128-bit, Grain, and Trivium for the performance characteristics: time to encrypt and decrypt; throughput of the encryption; the avalanche effect; and results of statistical testing. The size of the data sets used for the experiments were: 2^6 bytes, 2^{10} bytes, 2^{12} bytes, and 21^5 bytes respectively. We have executed a cipher 50 times and then compared the average results.

A. Encryption/Decryption Time

The time required to convert plaintext into ciphertext is known as encryption time, whereas the time taken to transform ciphertext back into plaintext is referred to as decryption time. We have provided details about the time taken for cryptography and decryption processes of our proposed work along with other algorithms. Four sets of data differing in size were used and set encryption/decryption times were measured over fifty iterations. From there, averages were taken and compared with the other works and the proposed algorithm does indeed work as intended and performs with greater efficiency. The results are depicted in figures 3 and 4 where the encryption/decryption fluctuate.

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B. Throughput

One of the metrics of an algorithm, the throughput is also one measure of its efficiency. The throughput of the algorithm is typically associated with the performance of the system; that is, automatically the more performance-competent is the effective algorithm, the higher is the throughput.

The encryption/decryption throughput of our suggested work is computed in this section, and it is compared with the current technique. The results show that our cipher has a higher throughput than Grain, Grain-128, and Trivium. Additionally, the comparison result with the other methods is shown in Figs. 5 and 6.

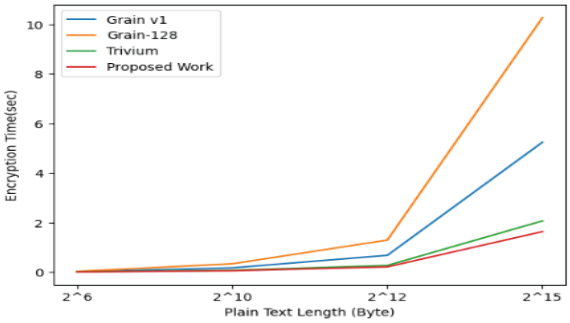


Fig. 3 Encryption time(sec)

C. Security Analysis

This part covers the method that identifies the randomness of the key using statistical means.

C.1 NIST test

The standard statistical test suites, NIST, are used to assess our stream encryption since statistical tests are essential for identifying statistical faults in a particular data collection. For the test, a sample size of 1500000 bits was used. Our stream cipher successfully passed the NIST test.

C.2 Avalanche Analysis

A cryptographic feature known as the "avalanche effect" quantifies the effect of altering an input parameter by one bit on the output resulting from various stream cipher stages. We measure the avalanche effect of altering one bit in the key to construct a keystream. An example of quantifying the avalanche effect of a single bit change in the key and its impact on the generated keystream. This scenario indicates that changing only one bit in the key results in an average change of about 53.118375% of the bits. We have compared the avalanche effect of different algorithms with our proposed work, as illustrated in Fig.7.

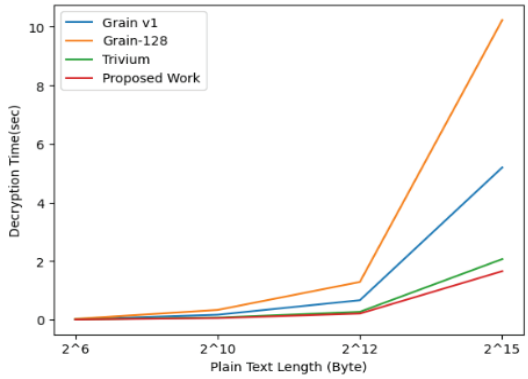


Fig.4. Decryption Time(sec)

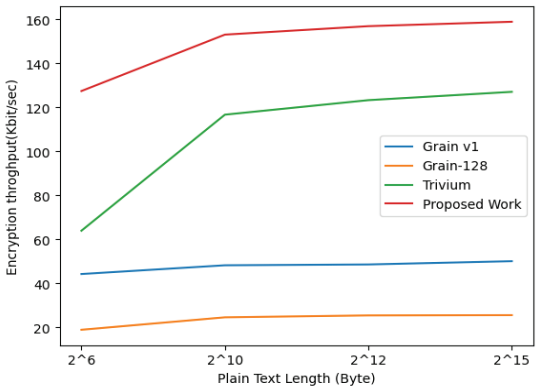


Fig.5 Encryption throughput

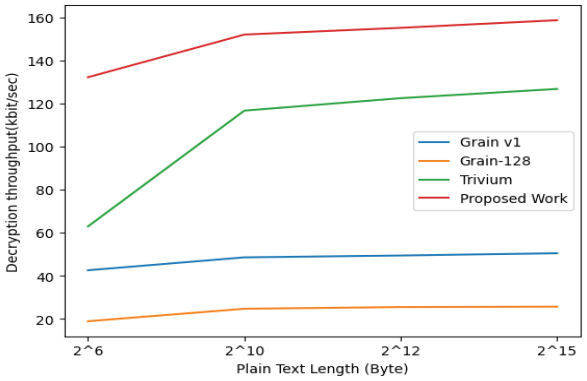


Fig. 6 Decryption throughput(kbit/sec)

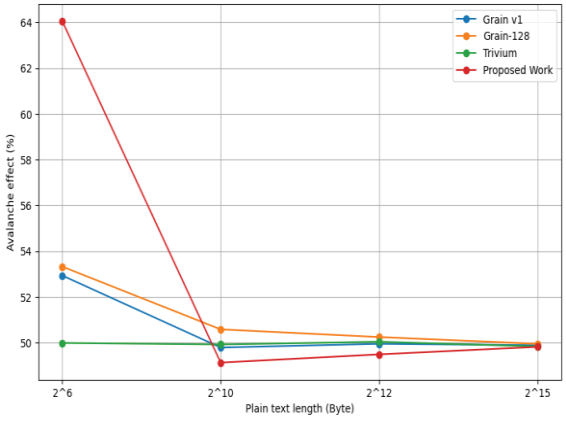


Fig.7 Avalanche effect (%)

VI. Observations

This section explains how the proposed algorithms perform in terms of throughput, avalanche effect, encryption and decryption time, and their ability to successfully pass each of the NIST statistical tests individually. The first thing that our work proposes is the implementation of two NFSRs and a logistic chaos function. The combination of 3 non-linear combinations produces randomness in the keystream, and as a result, all 16 NIST tests and the Avalanche effect are passed. The second part is that we have two 40-bit NFSR, which are much lower than Grain, Grain-128, and Trivium. Therefore, the throughput is quite high, and the processing time is very low.

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VII. Conclusion and Future Direction

The development of lightweight cryptographic algorithms has been driven by the emergence of several interesting lightweight applications in academia and industry. The security and performance levels should be balanced by the lightweight cryptographic technique. Our research presents a lightweight stream cipher that achieves a balance between performance and security. Our cipher performs better in every way, including encryption/decryption time, throughput, NIST test, and avalanche effect, as we have shown in this study and compared it to several state-of-the-art algorithms, including Trivium, Grain, and Grain-128. Because of the results, we have discovered that the processing time has decreased to around 80% from Grain, 90% from Grain-128, and 50% from Trivium, while the throughput has increased up to approximately four times from Grain-128, three times from Grain, and one-third times from Trivium. It will be extremely tough for the attackers to predict the key because of the chaos and the two NLFSRs working together to produce more random keys. Due to its highly lightweight design, which compromises performance and security,The proposed scheme can be implemented on resource-constrained devices that transmit sensitive information to the cloud. Furthermore, the proposed approach can also be extended to applications such as video, audio, and image encryption.

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Data availability: The dataset can be accessed at .https://www.kaggle.com/datasets/crawford/20-newsgroups

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